

10 Chapters · 400+ Questions · 3 Quiz Levels · Free Access · No Account Required

COURSE OVERVIEW — CHAPTER BY CHAPTER

- 1. Introduction** — Before IEC 61131-3, every PLC manufacturer used a proprietary language. Engineers retrained from scratch when they changed vendors. This chapter explains why the standard exists, what it unified, and which of the five languages to reach for first.
- 2. Data Types** — BOOL, INT, REAL, TIME, STRING, and user-defined structured types. The type system is what separates safe PLC code from implicit conversions that silently corrupt process values at runtime.
- 3. Structured Text (ST)** — Pascal-style text language: IF/THEN, FOR, WHILE, CASE, function calls. The most expressive of the five languages and the most transferable skill for engineers coming from software backgrounds.
- 4. Ladder Diagram (LD)** — Relay-logic graphical representation: contacts, coils, and rungs. Electricians can read it without training. Every PLC vendor supports it. Understanding LD is non-optional for anyone working with existing automation code.
- 5. Function Block Diagram (FBD)** — Signal-flow graphical language where data passes between interconnected function blocks. The natural fit for control engineers working with PID loops, signal conditioning, and process instrumentation.
- 6. Sequential Function Chart (SFC)** — State machine language for batch and sequential processes. Steps, transitions, and divergence structures replace thousands of lines of LD timer-and-flag logic with a readable, maintainable flowchart.
- 7. Program Organization Units** — Functions, Function Blocks, Programs, and Configurations are the module system that enables reusable automation libraries. Engineers learn how to build code that works across vendors without modification.
- 8. RTAC Integration** — Applying IEC 61131-3 logic in the SEL RTAC; how the standard crosses from PLC programming into substation protection and automation. The bridge between this course and the RTAC guide.
- 9. Troubleshooting** — Type mismatches, scan cycle overruns, unintended coil latching, and the runtime faults that crash programs without a clear error message. Field debugging procedures for real PLC hardware.
- 10. Practice Lab** — Four exercises: a PID temperature controller in Structured Text, a motor starter sequence in Ladder, a batch filling process in SFC, and a reusable Function Block library for analog scaling.

Certification Relevance

- **PLCopen Certified** — PLCopen certification exams test IEC 61131-3 language knowledge, POU, and programming best practices directly.
- **ISA CCST** — PLC programming, ladder logic, and control system configuration are core tested topics at all levels.
- **Vendor Credentials** — Siemens TIA Portal, Rockwell Studio 5000, and Beckhoff TwinCAT all use IEC 61131-3 as their programming foundation.

Next Steps

1. Complete IEC 61131-3 before starting the RTAC guide — logic concepts transfer directly to ACCELERATOR Architect.
2. Download CODESYS (free IDE, IEC 61131-3 compliant) and run the practice lab exercises in a soft PLC.
3. Review the PLCopen Function Block libraries for motion, communications, and safety.
4. Pursue PLCopen certification or a vendor credential (TIA Portal, Studio 5000) as the first PLC-specific credential.